

PHYS 225

Week 1, Lecture 1:
Introduction to Special Relativity

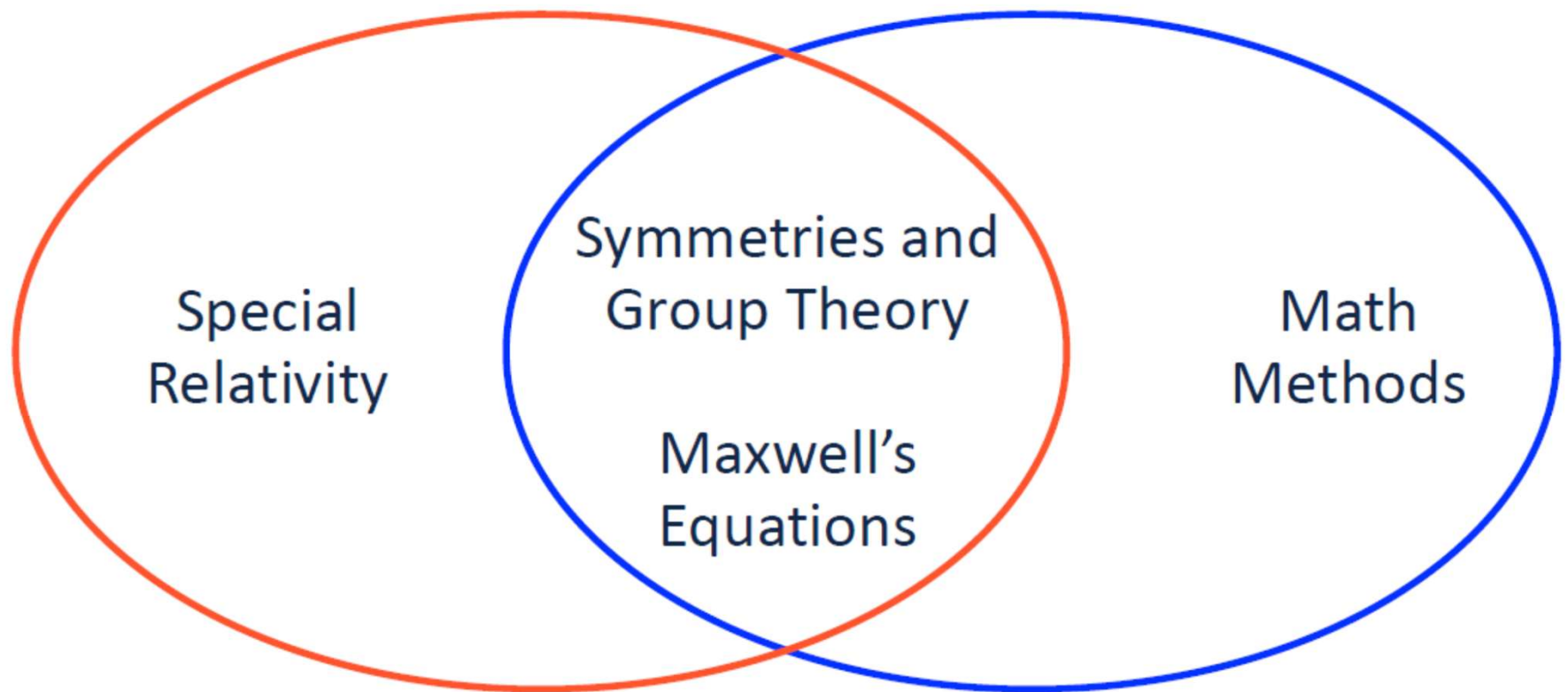


UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

PHYS 225 ©2022-2026 UIUC. All rights reserved.

Welcome to PHYS 225!

Plan for this course:



Course Staff

- **Lecturer:** Dr. Elena Koptieva
- **Discussion TAs:** Io Kovach
Freddy Pardoe
Michael Wentzel
Sami Muslih
- **Graders:** Hanqiao Cheng
Zoey Dong
Kittiya Plianaek
Max Ma
Spyros Thomopoulos
Ivan Wang
Nan Zhang

For further contact information see the course website:

<https://courses.physics.illinois.edu/phys225/sp2026/contact-info.html>

About me

Witek gravity group

witek-gravity-group.physics.illinois.edu

Personal webpage

elena-kopteva.github.io

$$\begin{aligned} \nabla_\mu \nabla^\mu &= (\nabla_\mu - iq A_\mu)(\nabla^\mu - iq A^\mu) = \nabla_\mu \nabla^\mu + (-1)q^2 A_\mu A^\mu - iq A_\mu \nabla^\mu - iq \nabla^\mu A_\mu \\ &= \nabla_\mu \nabla^\mu - q^2 A_\mu A^\mu - iq A_\mu \nabla^\mu - iq \nabla^\mu A_\mu \\ &= \nabla_\mu \nabla^\mu - q^2 A_\mu A^\mu - iq (A_\mu \nabla^\mu + A^\mu \nabla_\mu) \end{aligned}$$

0 due to Lorenz gauge

$$\left\{ A^\mu \nabla_\mu = g^{\mu\nu} A_\mu \nabla_\nu = A_\mu \nabla^\mu \right\}$$

$$\nabla_\mu \nabla^\mu = \nabla_\mu \nabla^\mu - 2iq A_\mu \nabla^\mu - q^2 A_\mu A^\mu$$

$$\nabla_\mu \nabla^\mu = \nabla_\mu \nabla^\mu - 2iq A_\mu \frac{\partial}{\partial t} - q^2 A_\mu A^\mu$$

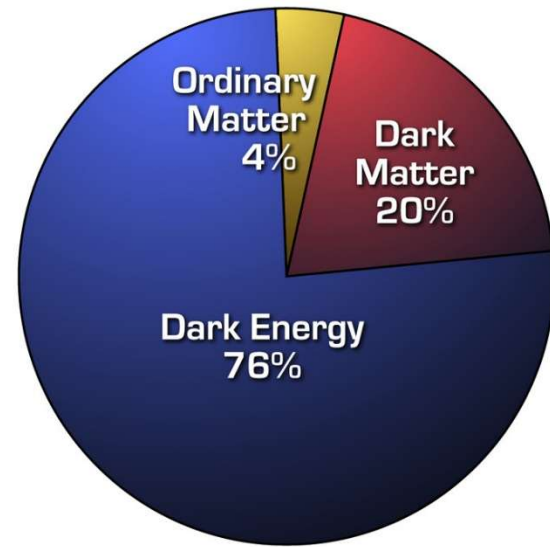
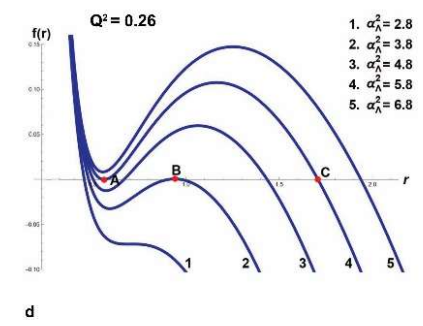
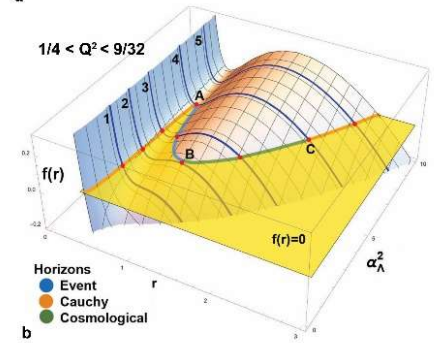
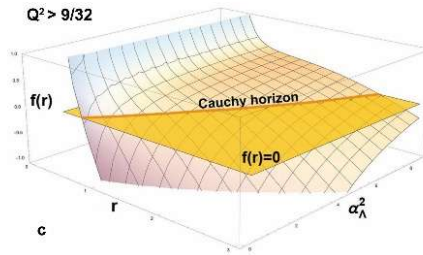
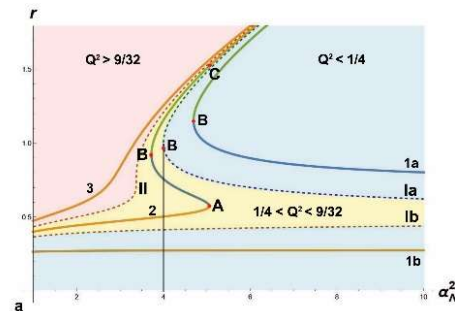
$$\boxed{\nabla_\mu \nabla^\mu = \nabla_\mu \nabla^\mu - 2iq \left(-\frac{Q}{r}\right) \frac{\partial}{\partial t} + \frac{q^2 Q^2}{f(r)r^2}}$$

$$A_\mu = \left(-\frac{Q}{r}, 0, 0, 0\right) \Rightarrow$$

$$\Rightarrow A_\mu \nabla^\mu = A_0 \nabla^0$$

$$A_0 \nabla^0 = A_0 A^0$$

$$A^0 = g^{00} A_0 = -\frac{1}{f(r)} \cdot \left(-\frac{Q}{r}\right) = \frac{Q}{f(r)}$$



Communication

- **Step 1:** Refer to the **course website**
<https://courses.physics.illinois.edu/phys225/sp2026/index.html>
- **Step 2:** Come to Office Hours:
Wednesdays 4-7 pm, Loomis 276
Thursdays 5-7 pm, Loomis 297
- **Step 3:** Email your Discussion TA
(response time not guaranteed)

Grading

20% Lecture attendance and participation (1-minute papers)

20% Homework (weekly)

10% Discussions

50% Exams (midterm and final)

Bonus does exist!

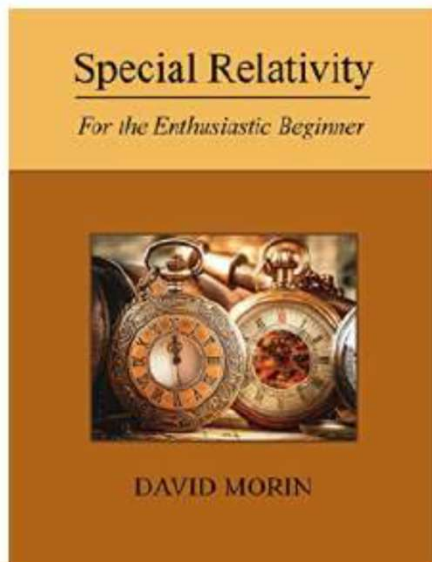
Everything except exams submitted through **Gradescope**.

Sign up now if you haven't already! Code **6XKGKN**

See **course website** for more details

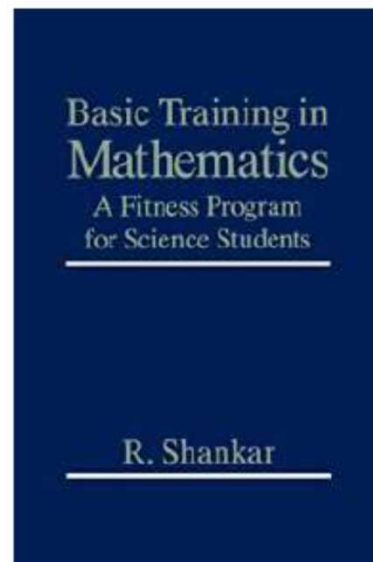
Textbooks

Required:



\$5.95 as a Kindle e-book on Amazon,
also in bookstore

Optional:



In bookstore (and on Amazon)

Readings are expected to be done **before** class
(optional readings in parentheses)

Upcoming Dates

- **Discussion** starts this week, Friday 1/23, 2026
- **Homework 1:**
 - Opens Thursday 1/22, 2026
 - Due Thursday 1/29, 2026
- **Office Hours:**
 - Start next week
 - Wednesdays 4-7 pm, Loomis 276
 - Thursdays 5-7 pm, Loomis 297
- **Midterm Exam**
 - 8th week, 3/10, 2026, in class

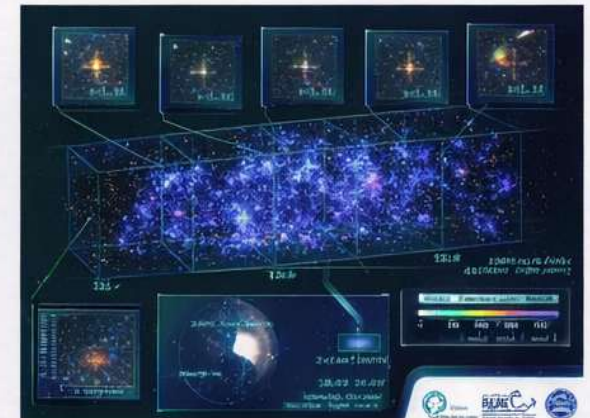
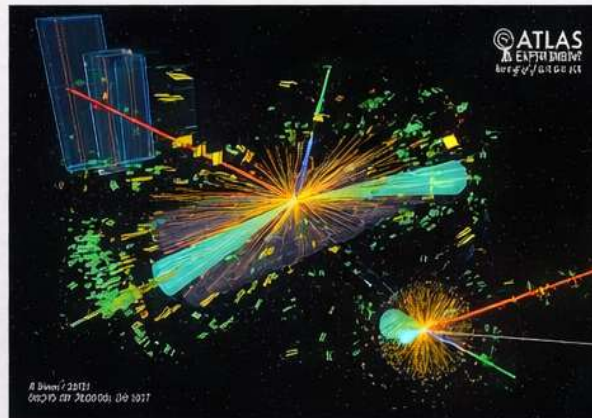
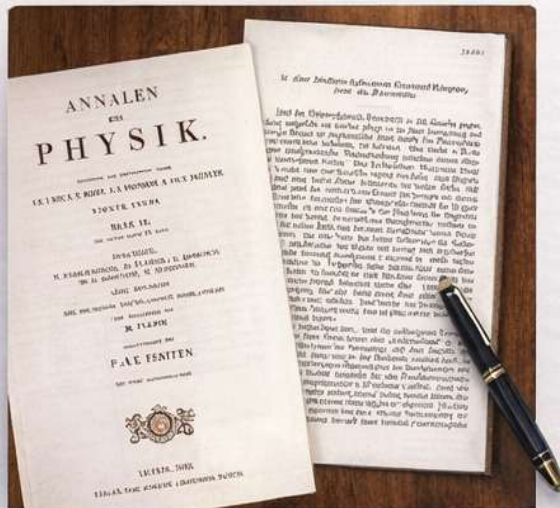
A word about discussions and HW

- This course is the gateway to upper-level physics classes.
- In this course (and from now on), the lecture and discussion will not “tell you how to solve the problems.” They will give you all the tools you need, but the point of the homework is to wrestle with your understanding and make sure you have internalized the material.
- The goal of discussion sections is to **enrich and deepen** your knowledge of the material. If you need input on how it all fits together, **ask!**
- If you're stuck on HW:
 - Try the worked problems in the textbook! There are LOTS.
 - Start EARLY. Panic and scrambling are not good problem-solving tools!
 - **Work with your friends and classmates!**
 - Come to office hours, but try the HW yourself first.

Week 1, Lecture 1

Introduction to Special Relativity

The theory of special relativity was first written down in 1905, and has passed every experimental test we've thrown at it over the last 100 years, from subatomic particles to galaxies

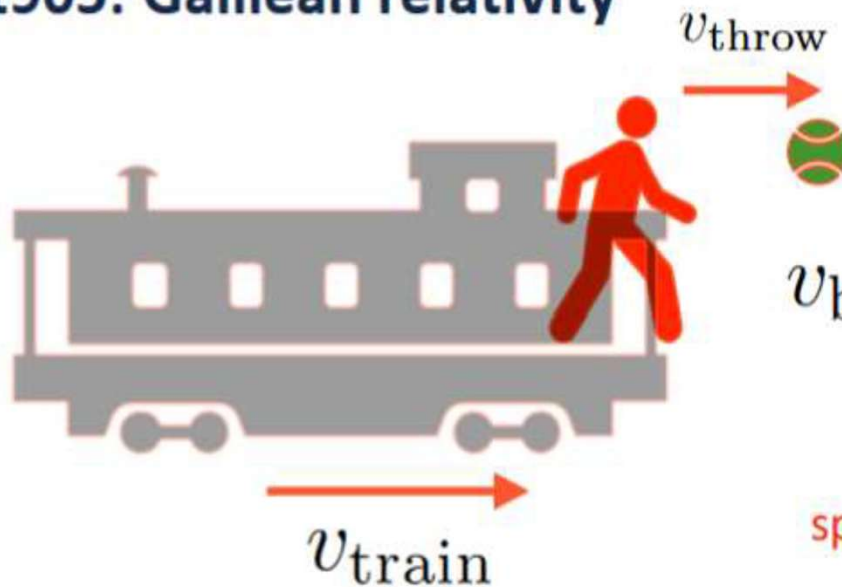


One Minute Paper: Write down one topic that you are excited to learn about in PHYS 225.

Week 1, Lecture 1

Introduction to Special Relativity

Pre-1905: Galilean relativity



$$v_{\text{ball}} = v_{\text{train}} + v_{\text{throw}}$$

speed of light: $c \approx 3 \times 10^8 \text{ m/s}$

Post-1905: special relativity

$$v_{\text{ball}} = \frac{v_{\text{train}} + v_{\text{throw}}}{1 + v_{\text{train}}v_{\text{throw}}/c^2}$$

In discussion, you will show that this is a speed limit: **can't exceed c !**

Week 1, Lecture 1

Introduction to Special Relativity

Galilean relativity:

1. Newton's Laws of motion are the same in all inertial reference frames (constant velocity)
2. Time is universal and the same for all observers



Isaac Newton

“Absolute, true and mathematical time, of itself, and from its own nature flows equably **without regard to anything external...**”



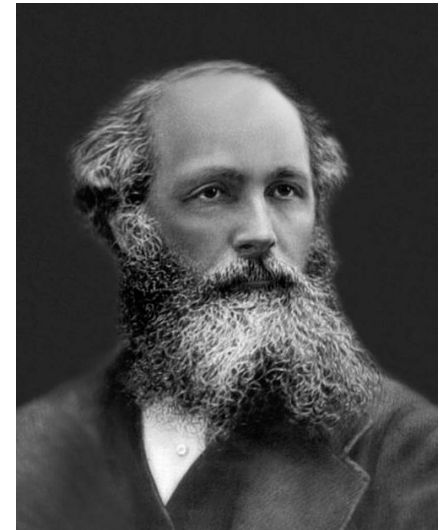
Galileo Galilei

“Have the ship proceed with any speed you like, so long as the motion is uniform and not fluctuating this way and that. You will discover not the least change in all the effects named, **nor could you tell from any of them whether the ship was moving or standing still.**”

Week 1, Lecture 1

Introduction to Special Relativity

Special Relativity began with a crisis in physics around 1900, as **Maxwell's equations** predicted a constant speed of light, clashing with **Newtonian mechanics**, leading to experiments like the **Michelson-Morley experiment** failing to find the 'aether'.



James Maxwell

Maxwell's Equations

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

Gauss's Law

$$\nabla \cdot \mathbf{B} = 0$$

Gauss's Law for Magnetism

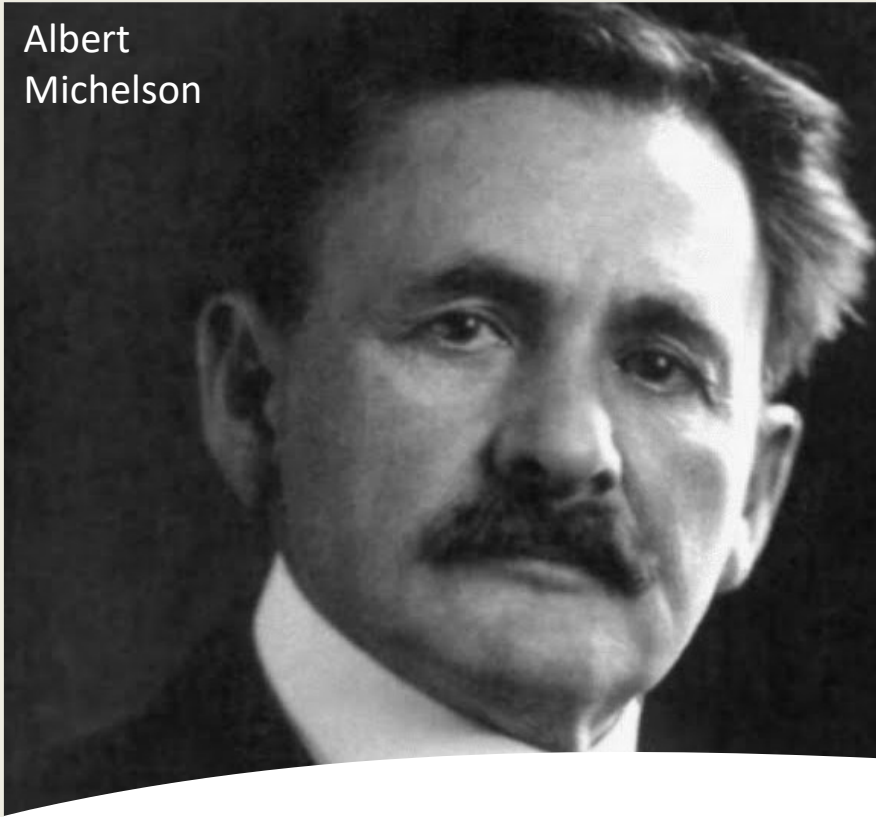
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Faraday's Law

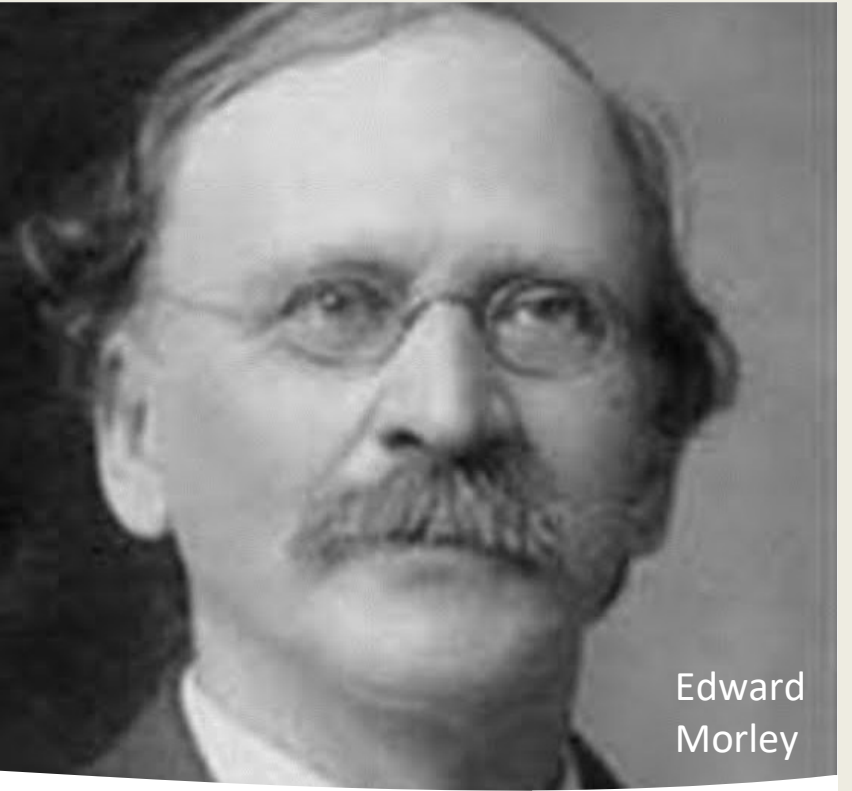
$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

Ampere's Law

Albert
Michelson



Edward
Morley



Michelson- Morley experiment

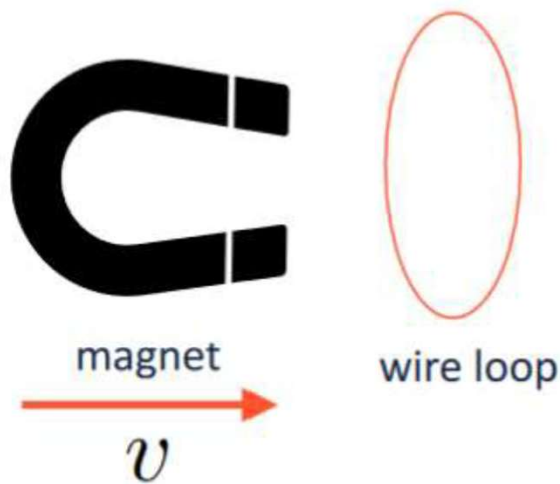
The null results from Michelson-Morley's experiment in 1887 initiated a line of research that eventually led to Einstein's theory of Special Relativity.

Week 1, Lecture 1

Introduction to Special Relativity

But when you include electromagnetism, there is a problem...

If the **magnet** moves, changing flux induces an **electric field**:



$$\int \mathbf{E}_{\text{loop}} \cdot d\mathbf{l} = - \frac{d\Phi_{\text{magnet}}}{dt}$$

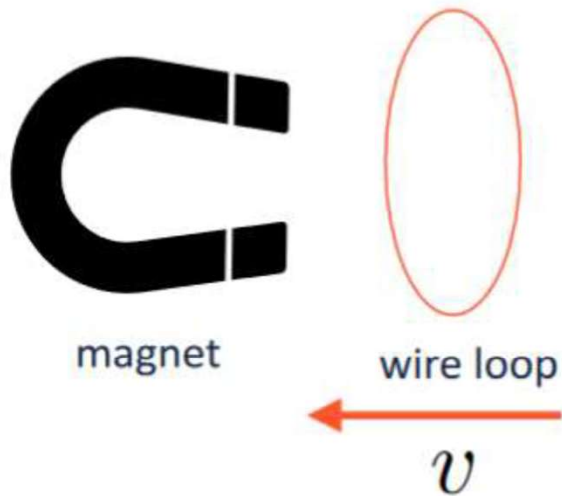
Week 1, Lecture 1

Introduction to Special Relativity

But when you include electromagnetism, there is a problem...

If the **loop** moves, the charges in the loop experience a **Lorentz force**:

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$$



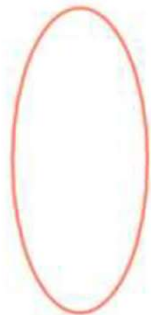
Week 1, Lecture 1

Introduction to Special Relativity

But when you include electromagnetism, there is a problem...



magnet



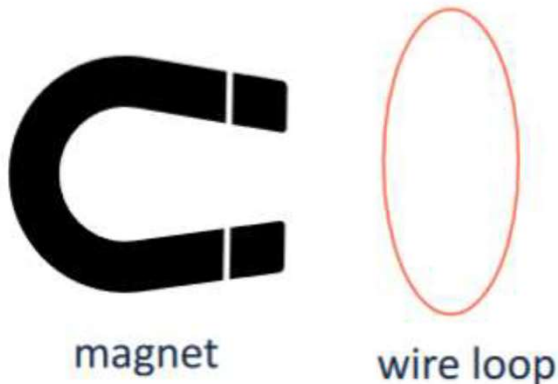
wire loop

Electric fields and Lorentz forces look like totally different things, but “miraculously”, the net force on the loop is the same in both frames!

Week 1, Lecture 1

Introduction to Special Relativity

But when you include electromagnetism, there is a problem...



Electric fields and Lorentz forces look like totally different things, but “miraculously”, the net force on the loop is the same in both frames!

Albert Einstein resolved this conflict in 1905 with his paper "On the Electrodynamics of Moving Bodies," proposing that the **speed of light is constant** for all observers and that **space and time are relative, not absolute**, reconciling electromagnetism and mechanics.

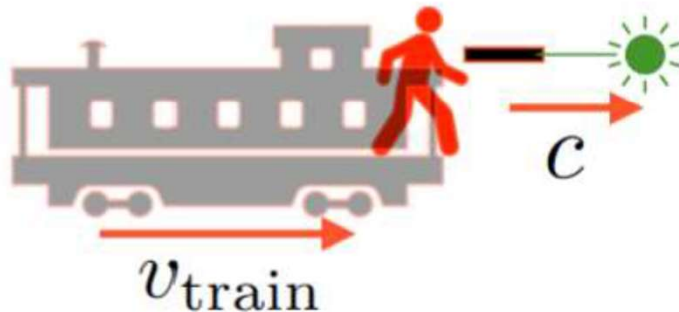
Week 1, Lecture 1

Introduction to Special Relativity

Postulates of Einstein's theory of special relativity:

1. ~~Newton's Laws of motion~~ All laws of physics are the same in all inertial reference frames (constant velocity)
2. ~~Time is universal and the same for all observers~~ The speed of light in vacuum is constant in all inertial frames

(The "special" in "special relativity" means this theory describes inertial frames: the general theory of relativity can describe arbitrarily accelerating observers)



Watching the train go by:

~~$$v_{\text{laser}} = c + v_{\text{train}}$$~~

$$v_{\text{laser}} = c$$

In discussion, you will show that postulate 2 implies that time is **not** the same in different inertial frames!

Week 1, Lecture 1

Introduction to Special Relativity

Summary

- The speed of light in vacuum is constant in all inertial frames
- Newton and Galileo thought time is absolute and universal, but that's not how our world behaves!
- Velocity addition is different when the speed of light is constant:

$$v_{AB} = \frac{v_A + v_B}{1 + v_A v_B / c^2}$$

(we will derive this formula in week 5)

Week 1, Lecture 1

Introduction to Special Relativity

Wrap-up

- Turn in your 1-minute papers **today** by 5 :59pm
- This week's reading: Morin 1.1-1.2, Appendix F 6.6.1-6.6.3
- Next week's reading: Morin 1.3-1.4
- Ask questions in office hours next week!
- Homework 1 assigned Thursday, due next Thursday

One Minute Paper: what does the “special” refer to in “special relativity”?